Title: Does Squad Age Decide World Cup Fate? A Statistical Analysis of FIFA World Cup (1930–2022)

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Introduction

FIFA World Cup is probably the biggest thing in football. Every four years countries send their best 23 players to compete, and coaches have to make tough decisions about who to pick. One question that always comes up is whether it's better to go with experienced older players or young energetic ones.

I got curious about this and decided to actually look at the data. This project uses World Cup data from **1930 all the way to 2022 covering 22 tournaments, nearly 14,000 player records** to see if squad age actually makes a difference in how teams perform.

I tried to answer four things:

1. Does a team's average age affect how many goals they score?
2. Does age have anything to do with reaching the knockout stage?
3. Do teams from different parts of the world differ in squad age?
4. And what do actual football fans think about all this?

Data

Four datasets were loaded from the **Fjelstul World Cup Database** — an open source database built by researcher Joshua Fjelstul covering all men's World Cups from 1930 to 2022.

squads (13,843 rows) — every player in every World Cup squad

players (10,401 rows) — player details including birth dates

tournaments (30 rows) — tournament results, host countries, winners

matches (1,248 rows) — all match scores

The birth date column in the players dataset was the key variable needed for this project. It was later joined with the squads data to calculate each player's age at the time of their World Cup.

```r
# Phase 1: Load datasets

library(tidyverse)

library(lubridate)

squads      <- read_csv("https://raw.githubusercontent.com/jfjelstul/worldcup
/master/data-csv/squads.csv")

players     <- read_csv("https://raw.githubusercontent.com/jfjelstul/worldcup
/master/data-csv/players.csv")

tournaments <- read_csv("https://raw.githubusercontent.com/jfjelstul/worldcup
/master/data-csv/tournaments.csv")

matches     <- read_csv("https://raw.githubusercontent.com/jfjelstul/worldcup
/master/data-csv/matches.csv")


# Phase 2: Data Cleaning

# Filter to men's only
tournaments_men <- tournaments %>%
  filter(!str_detect(tournament_name, "Women"))

players_men <- players %>%
  filter(female == 0)


# Join everything together
squads_full <- squads %>%
  left_join(players_men %>% select(player_id, birth_date),
            by = "player_id") %>%
  left_join(tournaments_men %>% select(tournament_id, year, start_date, winne
r),
            by = "tournament_id")
```

```r
# Calculate player ages
squads_full <- squads_full %>%
  mutate(
    birth_date = as.Date(birth_date),
    age = year - year(birth_date)
  )


# Remove bad/missing ages
squads_full <- squads_full %>%
  filter(!is.na(age)) %>%
  filter(age >= 15, age <= 50)

summary(squads_full$age)
```

```
   Min.  1st Qu.  Median   Mean  3rd Qu.    Max.
  17.00   24.00   27.00   27.02   30.00   45.00
```

```r
nrow(squads_full)
```

```
 [1] 10893
```

```r
# Building squad summary table
squad_summary <- squads_full %>%
  group_by(tournament_id, year, team_name) %>%
  summarise(
    avg_age    = mean(age),
    age_sd     = sd(age),
    min_age    = min(age),
    max_age    = max(age),
    squad_size = n(),
    .groups    = "drop"
  )

# Adding knockout variables
home_knockout <- matches %>%
  filter(knockout_stage == 1) %>%
  select(tournament_id, team_name = home_team_name) %>%
  mutate(reached_knockout = 1)

away_knockout <- matches %>%
  filter(knockout_stage == 1) %>%
  select(tournament_id, team_name = away_team_name) %>%
  mutate(reached_knockout = 1)

knockout_teams <- bind_rows(home_knockout, away_knockout) %>%
  distinct()
```

```r
squad_summary <- squad_summary %>%
  left_join(knockout_teams, by = c("tournament_id", "team_name")) %>%
  mutate(reached_knockout = ifelse(is.na(reached_knockout), 0, 1))

table(squad_summary$reached_knockout)
```

0	1
242	247

```r
head(squad_summary, 10)
```

A tibble: 10 × 9

	tournament_id	year	team_name	avg_age	age_sd	min_age	max_age	squad_size
1	WC-1930	1930	Argentina	24.9	3.11	20	31	22
2	WC-1930	1930	Belgium	25.5	5.01	19	38	16
3	WC-1930	1930	Bolivia	25	4.13	18	32	16
4	WC-1930	1930	Brazil	24.8	2.92	18	30	24
5	WC-1930	1930	Chile	26.4	3.20	20	33	19
6	WC-1930	1930	France	24.1	1.82	22	28	16
7	WC-1930	1930	Mexico	24.5	3.71	18	34	17
8	WC-1930	1930	Paraguay	24.5	3.21	20	29	6
9	WC-1930	1930	Peru	23.5	3.26	19	30	19
10	WC-1930	1930	Romania	23.3	3.10	19	30	15

```r
nrow(squad_summary)
```

```
[1] 489
```

Data Cleaning

The raw data needed quite a bit of work before it was ready for analysis. First, women's tournaments and female players were filtered out. The players dataset had both male and female records mixed together, identified by a female column **(0 = male, 1 = female)**.

After that, the four datasets were joined together into one single table using player_id and tournament_id as the common keys. Player ages were then calculated simply by subtracting birth year from tournament year.

A few data quality issues showed up .Some players had **negative ages or clearly wrong birth dates**. These were removed by keeping only players aged between 15 and 50. Missing birth dates from older tournaments (**mainly 1930s–1950s**) were also **dropped**, which reduced the dataset from **13,843 to 10,893** player records.

The final working table, **squad_summary , has 489 rows**, one row per team per tournament. Each row contains the team's **average age, age spread, youngest player, oldest player and squad size.**

```r
# Phase 3: Data Analysis

summary(squad_summary$avg_age)
```

```
Min.   1st Qu.  Median    Mean   3rd Qu.  Max.
22.06   26.23   27.05    26.99   27.91   30.27
```

```r
summary(squad_summary$age_sd)
```

```
Min.   1st Qu.  Median    Mean   3rd Qu.  Max.
1.786   3.158   3.496    3.568   3.960   6.404
```

<u>Exploratory Data Analysis</u>

The average squad age across all 22 World Cups is about 27 years, with a minimum of 22 and maximum of 30. This is a pretty narrow range , that means coaches across all eras have consistently preferred players in their mid to late twenties.

The age spread (standard deviation) averages around 3.6 years, meaning most squads carry a mix of players roughly between 23 and 31 years old.

```r
# Which tournament had the oldest and youngest squads on average?
squad_summary %>%
  group_by(year) %>%
  summarise(mean_age = mean(avg_age)) %>%
  arrange(desc(mean_age))
```

```
 A tibble: 22 × 2
      year   mean_age

   1  2018     27.9
   2  1998     27.6
   3  2002     27.5
   4  2006     27.4
   5  1994     27.4
   6  2010     27.4
   7  2014     27.3
   8  1986     27.1
   9  1982     27.1
  10  2022     27.0
# ℹ  12 more rows
```

```r
# Has average squad age changed over the decades?
squad_summary %>%
  group_by(year) %>%
  summarise(mean_age = mean(avg_age)) %>%
  print(n = 22)
```

```
# A tibble: 22 × 2
```

	Year	mean_age
1	1930	24.7
2	1934	25.5
3	1938	25.8
4	1950	26.9
5	1954	26.9
6	1958	26.8
7	1962	26.4
8	1966	26.3
9	1970	26.3
10	1974	26.8
11	1978	26.7
12	1982	27.1
13	1986	27.1
14	1990	26.9
15	1994	27.4
16	1998	27.6
17	2002	27.5
18	2006	27.4
19	2010	27.4
20	2014	27.3
21	2018	27.9
22	2022	27.0

Looking at the trend over the decades, squads have clearly gotten older. Back in **1930** the average age was just **24.7**. By **2018 it had climbed to 27.9**, nearly 3 years older over 90 years of football. Better fitness, longer careers and the increasing demands of modern football probably explain this. The slight drop in **2022 back to 27.0** is interesting , possibly because of exciting young players like Bellingham, Gavi and Pedri getting their chance.

```r
# Top 10 oldest squads
squad_summary %>%
  arrange(desc(avg_age)) %>%
  select(year, team_name, avg_age) %>%
  head(10)
```

```
# A tibble: 10 × 3

    year team_name           avg_age

  1  1998 Germany                30.3
  2  1954 South Korea            30
  3  1998 Scotland               30.0
  4  1998 Belgium                29.9
  5  1998 Austria                29.8
  6  2018 Costa Rica             29.7
  7  2006 Trinidad and Tobago    29.7
  8  2002 Belgium                29.5
  9  2018 Argentina              29.5
 10  1958 Sweden                 29.5


# Top 10 youngest squads
squad_summary %>%
  arrange(avg_age) %>%
  select(year, team_name, avg_age) %>%
  head(10)

 # A tibble: 10 × 3

    year team_name   avg_age

  1  1930 Yugoslavia     22.1
  2  1966 North Korea    22.9
  3  1930 Romania        23.3
  4  1930 Peru           23.5
  5  1998 Cameroon       23.7
  6  1954 Turkey         23.8
  7  1934 Argentina      23.9
  8  1958 Paraguay       24.0
  9  1938 Cuba           24
 10  2002 Nigeria        24.0
```

```r
# Add confederation manually
squad_summary <- squad_summary %>%
  mutate(confederation = case_when(
    team_name %in% c("Germany", "France", "Spain", "England",
                     "Italy", "Netherlands", "Portugal", "Belgium",
                     "Croatia", "Denmark", "Sweden", "Switzerland",
                     "Poland", "Serbia", "Czech Republic", "Hungary",
                     "Romania", "Scotland", "Turkey", "Ukraine",
                     "West Germany", "Soviet Union", "Yugoslavia",
                     "Czechoslovakia", "East Germany", "Austria",
                     "Bulgaria", "Norway", "Slovakia", "Slovenia",
                     "Wales", "Northern Ireland", "Greece", "Russia") ~ "UEFA
",
    team_name %in% c("Brazil", "Argentina", "Uruguay", "Colombia",
                     "Chile", "Paraguay", "Peru", "Ecuador",
                     "Bolivia") ~ "CONMEBOL",
    team_name %in% c("USA", "Mexico", "Costa Rica", "Honduras",
                     "El Salvador", "Cuba", "Canada",
                     "Trinidad and Tobago", "Haiti",
                     "Jamaica") ~ "CONCACAF",
    team_name %in% c("Nigeria", "Senegal", "Ghana", "Cameroon",
                     "Morocco", "Tunisia", "Ivory Coast", "Algeria",
                     "South Africa", "Egypt", "Togo", "Angola",
                     "Democratic Republic of the Congo",
                     "Zaire") ~ "CAF",
    team_name %in% c("Japan", "South Korea", "Australia",
                     "Saudi Arabia", "Iran", "Iraq", "China",
                     "North Korea", "UAE", "Kuwait",
                     "Indonesia") ~ "AFC",
    TRUE ~ "Other"
  ))

# Compare average age by confederation
squad_summary %>%
  group_by(confederation) %>%
  summarise(
    mean_age   = round(mean(avg_age), 2),
    teams      = n()
  ) %>%
  arrange(desc(mean_age))
```

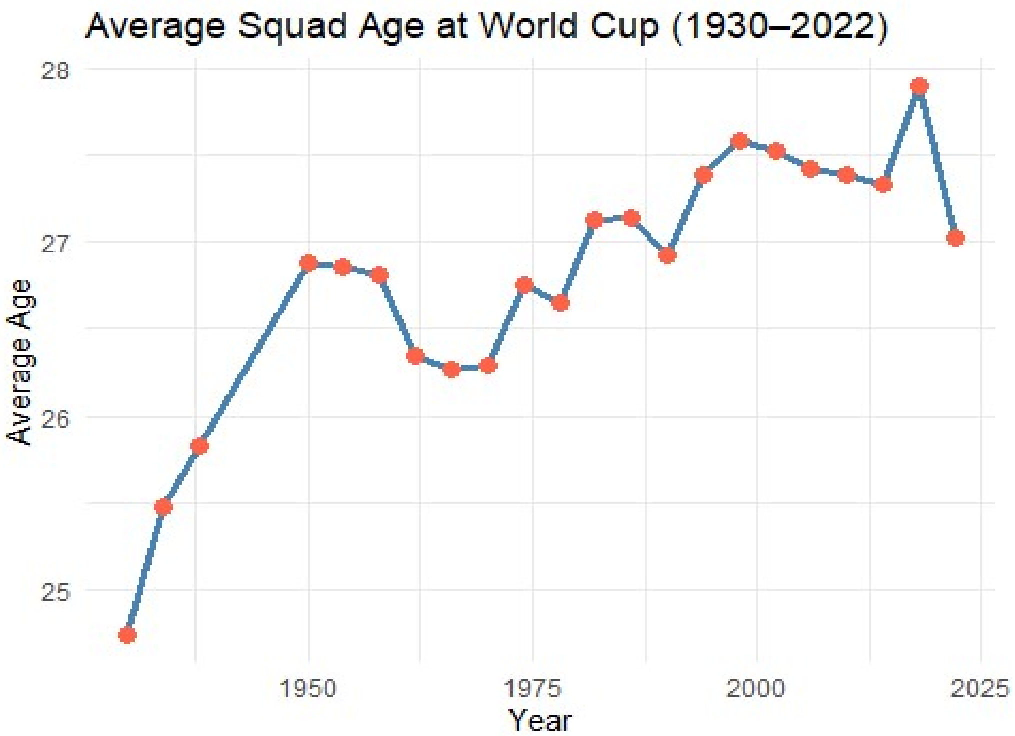
```
# A tibble: 6 × 3

  confederation   mean_age  teams

1 Other               27.3     24
2 CONCACAF            27.1     34
3 UEFA                27.1    252
4 AFC                 27.0     41
5 CONMEBOL            26.9     89
6 CAF                 26.4     49
```

Germany's 1998 squad holds the record for oldest ever at **30.3 years average**. What's striking is that four squads from the same 1998 tournament appear in the top 10 oldest list, that whole World Cup was dominated by veteran teams.

Yugoslavia 1930 was the youngest **at 22.1**. Most young squads are from the early tournaments which makes sense. **North Korea 1966 at 22.9 years** is a fun one though, that young team knocked out Italy which nobody expected.

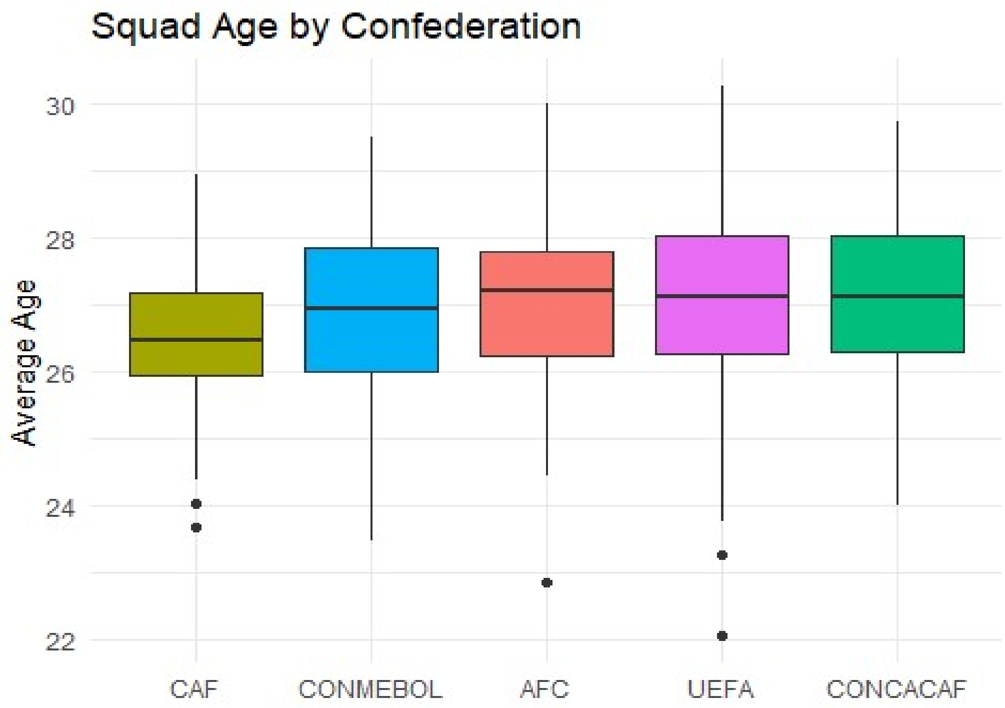
African teams average 26.4 years, **Europeans average 27.1**. Small difference but it shows different approaches to picking squads across regions.

```r
# Phase 4: Visualization
squad_summary %>%
  group_by(year) %>%
  summarise(mean_age = mean(avg_age)) %>%
  ggplot(aes(x = year, y = mean_age)) +
  geom_line(color = "steelblue", size = 1.2) +
  geom_point(color = "tomato", size = 3) +
  labs(title = "Average Squad Age at World Cup (1930-2022)",
       x = "Year", y = "Average Age") +
  theme_minimal()
```



The chart shows a pretty clear upward trend from 1930 to 2018. Squads went from averaging around 24 years to nearly 28. It's not a perfectly straight climb though — there are some dips in the 1960s and 70s before it picks up again. 2018 was the peak and 2022 dropped back a bit.

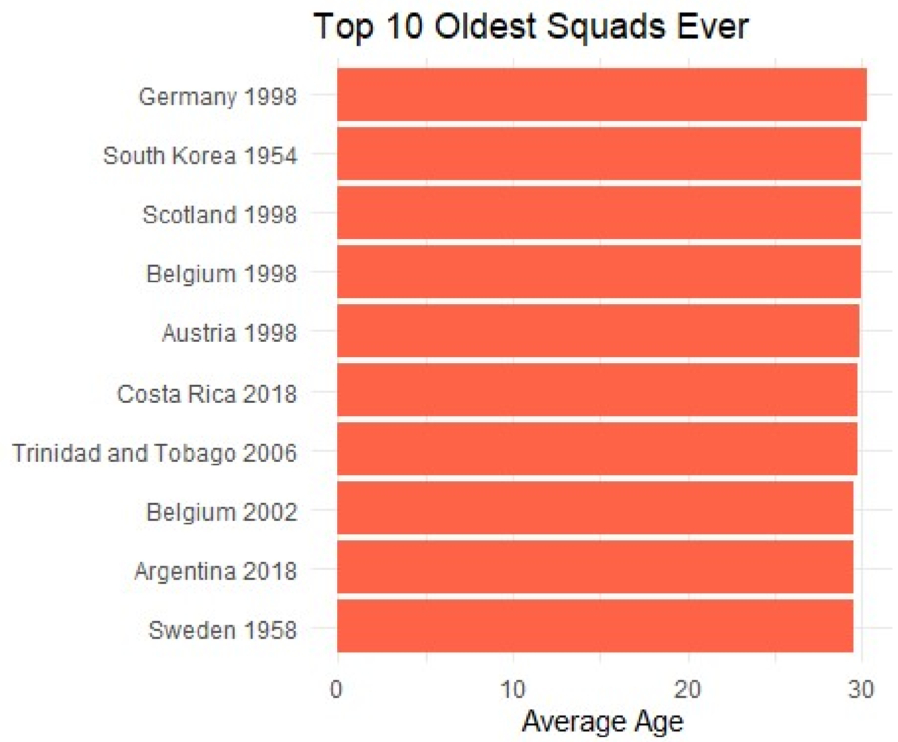
```
squad_summary %>%
  filter(confederation != "Other") %>%
  ggplot(aes(x = reorder(confederation, avg_age),
             y = avg_age, fill = confederation)) +
  geom_boxplot(show.legend = FALSE) +
  labs(title = "Squad Age by Confederation",
       x = "", y = "Average Age") +
  theme_minimal()
```



CAF sits the lowest which means African teams generally go with younger squads. **UEFA and CONCACAF** are at the top. The boxes overlap quite a bit though so the differences aren't massive visually.

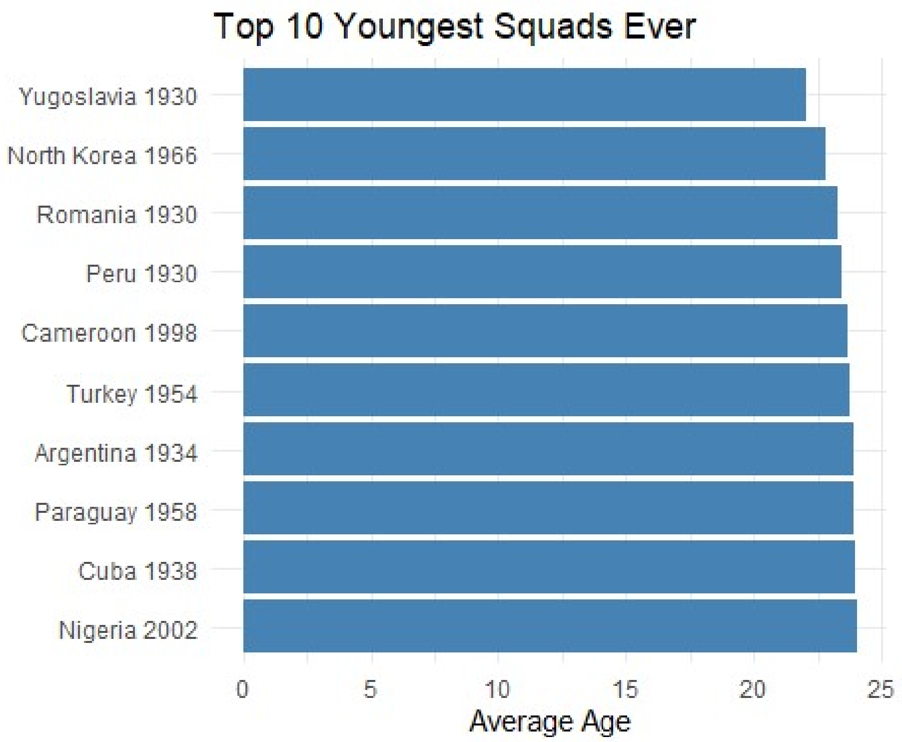
Those dots below the boxes are outliers, that is, extremely young squads, mostly from the early tournaments.

```
squad_summary %>%
  arrange(desc(avg_age)) %>%
  head(10) %>%
  mutate(label = paste(team_name, year)) %>%
  ggplot(aes(x = reorder(label, avg_age), y = avg_age)) +
  geom_col(fill = "tomato") +
  coord_flip() +
  labs(title = "Top 10 Oldest Squads Ever",
       x = "", y = "Average Age") +
  theme_minimal()
```



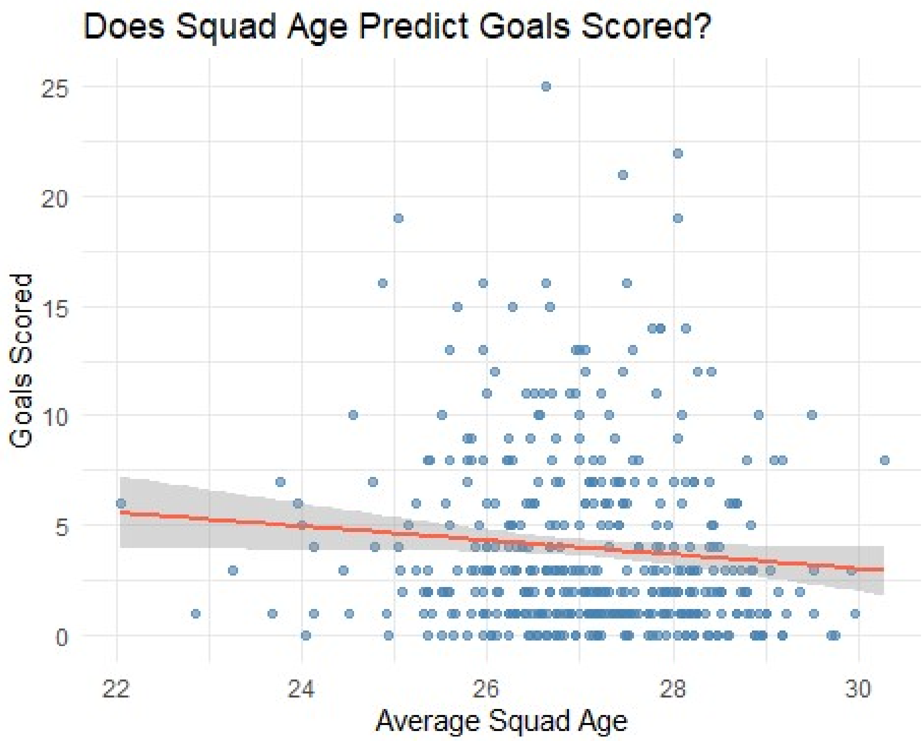
Germany **1998,more than 30 years** was the oldest squad ever. Four teams from that same 1998 tournament made this list which shows that whole World Cup was just full of experienced veteran squads. Argentina 2018 also shows up here that where Messi is the part of that squad. One of the most experienced teams in the tournament but they got knocked out by France in the Round of 16.

```
squad_summary %>%
  arrange(avg_age) %>%
  head(10) %>%
  mutate(label = paste(team_name, year)) %>%
  ggplot(aes(x = reorder(label, desc(avg_age)), y = avg_age)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Top 10 Youngest Squads Ever",
       x = "", y = "Average Age") +
  theme_minimal()
```



Yugoslavia 1930 at more than 22 years was the youngest squad ever recorded. Three teams from the 1930 tournament appear here which fits the overall age trend we saw earlier. North Korea 1966 is probably the most interesting entry as they knocked out Italy which is still considered one of the biggest upsets in World Cup history.

```
goals_data <- matches %>%
  group_by(tournament_id, home_team_name) %>%
  summarise(goals = sum(home_team_score, na.rm = TRUE), .groups = "drop") %>%
  rename(team_name = home_team_name)

squad_summary <- squad_summary %>%
  left_join(goals_data, by = c("tournament_id", "team_name"))

squad_summary %>%
  filter(!is.na(goals)) %>%
  ggplot(aes(x = avg_age, y = goals)) +
  geom_point(color = "steelblue", alpha = 0.6) +
  geom_smooth(method = "lm", color = "tomato") +
  labs(title = "Does Squad Age Predict Goals Scored?",
       x = "Average Squad Age", y = "Goals Scored") +
  theme_minimal()
```



The regression line slopes slightly downward. That means older squads tend to score a bit less goals. But the dots are scattered everywhere. So age clearly isn't the main factor in how many goals a team scores. Lots of other things matter more.

```
# Phase 5: Statistical Modeling
```

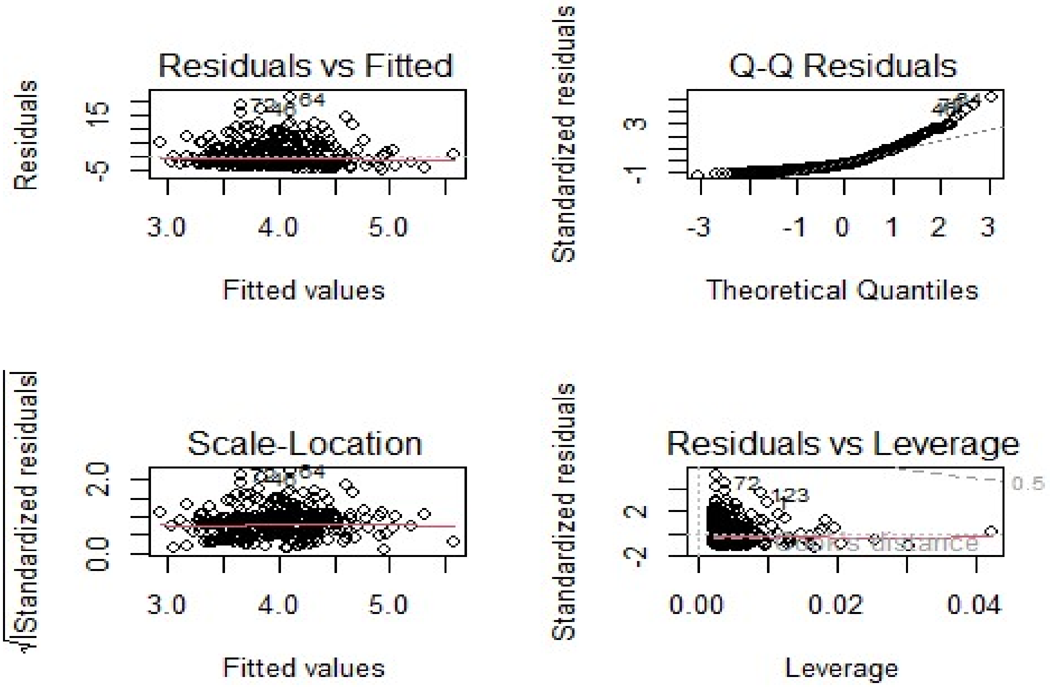
Three models were run to test whether squad age actually affects World Cup performance.

```
# Simple Linear Regression

model1 <- lm(goals ~ avg_age, data = squad_summary)
summary(model1)
```

```
Call:
lm(formula = goals ~ avg_age, data = squad_summary)

Residuals:
   Min      1Q    Median     3Q      Max
-4.938  -2.873   -1.389   1.944    20.897

Coefficients:
                 Estimate Std.     Error       t value     Pr(>|t|)
(Intercept)      12.6805           4.4307      2.862       0.00442 **
avg_age          -0.3220           0.1635      -1.970      0.04948 *
---

Residual standard error: 4.106 on 428 degrees of freedom
  (59 observations deleted due to missingness)
Multiple R-squared:  0.008987,   Adjusted R-squared:  0.006671
F-statistic: 3.881 on 1 and 428 DF,  p-value: 0.04948
```

Squad age is significant here (p = 0.049). As average age goes up by one year, goals scored drops by about 0.32. But R squared is only 0.009 so age explains less than 1% of the variation in goals. Significant but not practically very useful on its own.

```
# Model adequacy checking
par(mfrow = c(2, 2))
plot(model1)
```



Q-Q plot curves at the upper end meaning **residuals aren't perfectly normal**.

```
par(mfrow = c(1, 1))

shapiro.test(model1$residuals)


  Shapiro-Wilk normality test

 data:  model1$residuals
 W = 0.83536, p-value < 2.2e-16

library(lmtest)

bptest(model1)


  studentized Breusch-Pagan test

 data:  model1
 BP = 0.35493, df = 1, p-value = 0.5513

dwtest(model1)
```

```
 Durbin-Watson test

data:  model1
DW = 1.8979, p-value = 0.1425
alternative hypothesis: true autocorrelation is greater than 0
```

Shapiro-Wilk formally confirmed that **residuals are not normally distributed** ($p < 0.05$). This is expected though since goals scored is count data and naturally right skewed. The Breusch-Pagan test showed **no heteroscedasticity** ($p = 0.55$) and the Durbin-Watson statistic of 1.90 indicates **no autocorrelation** problem.

```r
# Logistic Regression

model2 <- glm(reached_knockout ~ avg_age,
              data   = squad_summary,
              family = binomial)

summary(model2)
```

```
Call:
glm(formula = reached_knockout ~ avg_age, family = binomial,
    data = squad_summary)

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.142800   1.923298  -0.074    0.941
avg_age      0.006049   0.071191   0.085    0.932


    Null deviance: 677.85  on 488  degrees of freedom
Residual deviance: 677.84  on 487  degrees of freedom
AIC: 681.84

Number of Fisher Scoring iterations: 3
```

```r
table(squad_summary$reached_knockout)
```

```
  0   1
242 247
```

The p-value is 0.932 which is about as far from significant as it gets. Squad age has essentially no effect on whether a team reaches the knockout stage.

```r
# ANOVA

model3 <- aov(avg_age ~ confederation,
              data = squad_summary %>%
                filter(confederation != "Other"))
summary(model3)
```

```
                Df Sum Sq Mean Sq F value Pr(>F)
 confederation   4   20.0   5.001   3.213 0.0128 *
 Residuals     460  716.2   1.557
 ---
```

The overall ANOVA result is significant **(p = 0.013)** meaning **at least one confederation is genuinely different from the others.**

```r
shapiro.test(model3$residuals)
```

```

  Shapiro-Wilk normality test

 data:  model3$residuals
 W = 0.99105, p-value = 0.006482
```

```r
#Levene's test for Homogeneity of variance
library(car)



leveneTest(avg_age ~ confederation,
           data = squad_summary %>%
             filter(confederation != "Other"))
```

```
 Warning in leveneTest.default(y = y, group = group, ...): group coerced to
 factor.

 Levene's Test for Homogeneity of Variance (center = median)
        Df F value Pr(>F)
 group   4  0.8147 0.5162
       460
```

Shapiro-Wilk gave **p = 0.006** but the W statistic is 0.991 which is extremely close to 1, so the normality violation is negligible. Levene's test confirmed equal variances across all confederations **(p = 0.516)**.

```r
TukeyHSD(model3)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = avg_age ~ confederation, data = squad_summary %>% filter
(confederation != "Other"))

$confederation
                          diff        lwr        upr     p adj
CAF-AFC            -0.62967874 -1.3529273 0.09356983 0.1214196
CONCACAF-AFC        0.08331336 -0.7092893 0.87591604 0.9984970
CONMEBOL-AFC       -0.17848191 -0.8234542 0.46649035 0.9423807
UEFA-AFC            0.03219922 -0.5432379 0.60763638 0.9998759
CONCACAF-CAF        0.71299210 -0.0497151 1.47569929 0.0796374
CONMEBOL-CAF        0.45119683 -0.1566621 1.05905574 0.2519555
UEFA-CAF            0.66187796  0.1283696 1.19538632 0.0066020
CONMEBOL-CONCACAF -0.26179527 -0.9507241 0.42713361 0.8362981
UEFA-CONCACAF      -0.05111414 -0.6754231 0.57319484 0.9994383
UEFA-CONMEBOL       0.21068113 -0.2106642 0.63202648 0.6476410
```

Out of all ten possible confederation pairs only UEFA vs CAF came out significant **(p = 0.007).** European squads are on average 0.66 years older than African squads. Every other combination showed no meaningful difference which suggests the age gap in world football is really just a Europe vs Africa story.

```r
# Non-parametric test for linear regression
cor.test(squad_summary$avg_age,
         squad_summary$goals,
         method = "spearman",
         use = "complete.obs")


  Spearman's rank correlation rho

data:  squad_summary$avg_age and squad_summary$goals
S = 14982336, p-value = 0.006669
alternative hypothesis: true rho is not equal to 0
sample estimates:
       rho
-0.1306489

# Non-parametric test for ANOVA
kruskal.test(avg_age ~ confederation,
             data = squad_summary %>%
               filter(confederation != "Other"))
```

```
	Kruskal-Wallis rank sum test

data:  avg_age by confederation
Kruskal-Wallis chi-squared = 13.682, df = 4, p-value = 0.008381
```

Since normality was violated in both the linear regression and ANOVA, non-para metric alternatives were also run as backup. Spearman correlation gave **p = 0.007** and Kruskal-Wallis gave **p = 0.008** both significant, same direction as the earlier models.

```
# Phase 6: Survey Analysis

survey <- read_csv("E:/World Cup Project/fifa survey.csv") %>%
  rename(
    follow_wc         = 1,
    experienced_win   = 2,
    youth_underrated  = 3,
    ideal_age         = 4,
    play_style        = 5,
    best_region       = 6,
    age_affects       = 7
  )
```

```
Rows: 18 Columns: 7
```

```
glimpse(survey)
```

```
Rows: 18
Columns: 7
$ follow_wc        <chr> "Very closely", "Casually", "Very closely", "Very c
lo…
$ experienced_win  <dbl> 1, 3, 3, 4, 3, 5, 4, 2, 3, 2, 5, 2, 3, 1, 4, 3, 4,
2
$ youth_underrated <dbl> 5, 4, 3, 4, 4, 4, 4, 1, 3, 4, 5, 3, 4, 4, 4, 4, 3,
3
$ ideal_age        <chr> "27 to 29", "27 to 29", "27 to 29", "27 to 29", "27
t…
$ play_style       <chr> "Balanced mix of youth and experience", "Balanced m
ix…
$ best_region      <chr> "Europe", "Europe", "South America", "Europe", "Eur
op…
$ age_affects      <dbl> 1, 2, 3, 4, 2, 4, 4, 4, 3, 4, 3, 4, 2, 4, 3, 4, 4,
2
```

```r
table(survey$follow_wc)
```

```
            Casually   Somewhat closely     Very closely 
                   5                  2               11 
```

Most respondents **(11 out of 18, or 61.1%)** reported following the FIFA World Cup very closely. Only five respondents followed it casually and two followed it somewhat closely. This suggests that the survey was completed primarily by football enthusiasts who are familiar with international football and World Cup tournaments. As a result, their opinions are likely informed by regular exposure to football competitions and player performances.

```r
table(survey$ideal_age)
```

```
27 to 29 
      18 
```

All respondents selected **27–29** years as the ideal average squad age for a World Cup team. This unanimous result is particularly interesting because the historical data analysis showed that the average World Cup squad age across all tournaments was approximately 27 years.

Respondents appear to believe that players in their late twenties represent the optimal balance between physical ability and competitive experience.

```r
table(survey$play_style)
```

```
Balanced mix of youth and experience             Experienced and tactical 
                                  13                                    4 
                 Young and energetic 
                                   1 
```

The majority of respondents **(72.2%)** preferred a balanced mixture of young and experienced players rather than relying heavily on either group. This finding suggests that football fans generally believe the most successful teams combine the energy, speed, and creativity of younger players with the leadership, tactical awareness, and composure of experienced players. The result aligns with modern squad-building strategies used by many successful national teams.

```r
table(survey$best_region)
```

```
          Europe       South America
           10             8
```

Opinions were divided between Europe and South America, with Europe receiving slightly more support (55.6%) than South America (44.4%). No respondent selected another region.

This reflects the historical dominance of European and South American nations in FIFA World Cup competitions. European teams have won many recent World Cups, while South American countries such as Brazil, Argentina, and Uruguay remain among the most successful football nations in history.

```r
summary(survey$experienced_win)
```

```
    Min.    1st Qu.   Median     Mean   3rd Qu.    Max.
       1        2        3        3        4        5
```

Responses regarding whether experienced squads perform better were relatively neutral. The mean and median values were **both 3 on the five-point scale**, indicating neither strong agreement nor strong disagreement.

This suggests that respondents do not believe experience alone guarantees success. While experience may provide advantages such as leadership and decision-making, fans appear to recognize that other factors such as talent, tactics, coaching quality, and team chemistry also play important roles.

```r
summary(survey$youth_underrated)
```

```
    Min.  1st Qu.  Median    Mean  3rd Qu.    Max.
   1.000   3.000   4.000   3.667   4.000   5.000
```

Respondents generally agreed that young squads are often underestimated. The average response was **3.67**, while the median response was **4**, indicating moderate agreement.

This perception may stem from several World Cup examples where younger teams exceeded expectations despite lacking experience. Respondents appear to believe that youthful squads can offer advantages such as speed, energy, adaptability, and motivation that are not always fully appreciated before tournaments begin.

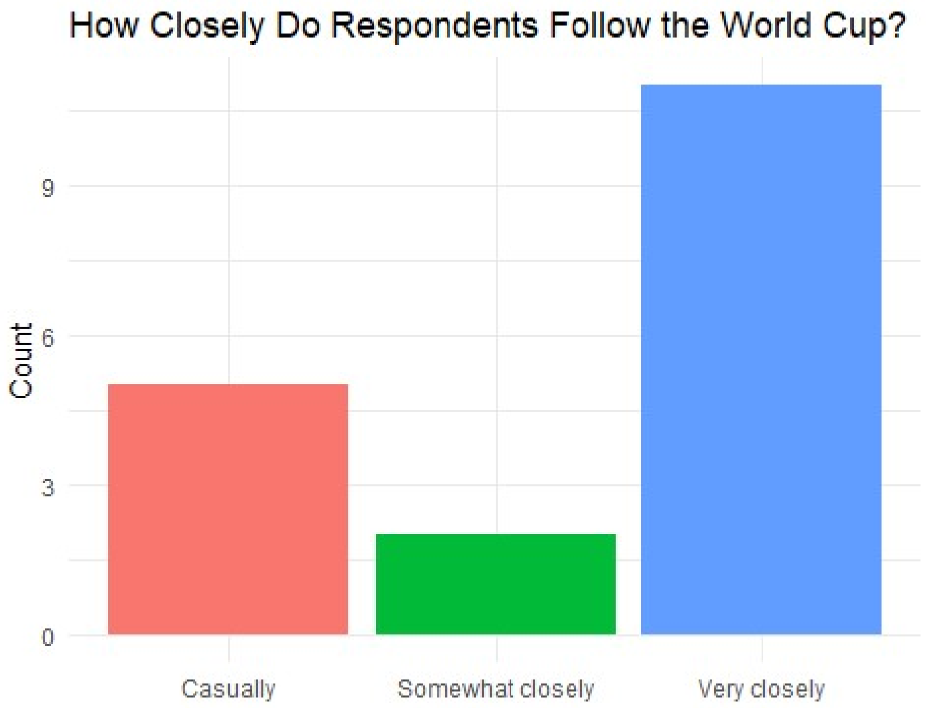
```r
summary(survey$age_affects)
```

```
    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
   1.000   2.250   3.500   3.167   4.000   4.000
```

Responses indicate a mild belief that squad age influences World Cup performance. The average score was slightly above neutral at **3.17**, and the median was **3.5**.

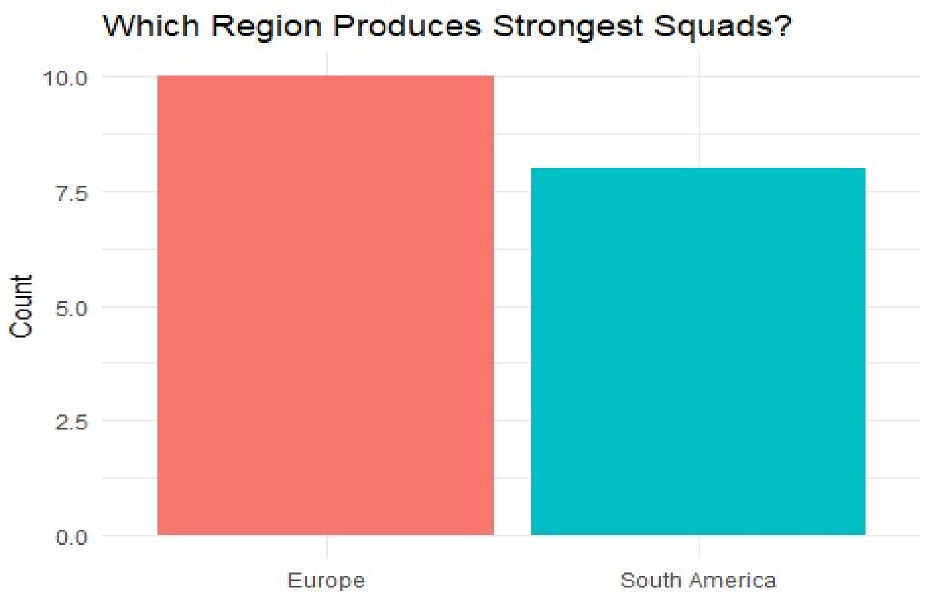
This suggests that respondents think age has some impact on team performance but is not the dominant factor determining success. The survey findings closely match the statistical analysis conducted earlier in the project, which found only weak relationships between squad age and actual World Cup outcomes.

```r
# Chart 1 - How closely do respondents follow WC?
ggplot(survey, aes(x = follow_wc, fill = follow_wc)) +
  geom_bar(show.legend = FALSE) +
  labs(title = "How Closely Do Respondents Follow the World Cup?",
       x = "", y = "Count") +
  theme_minimal()
```

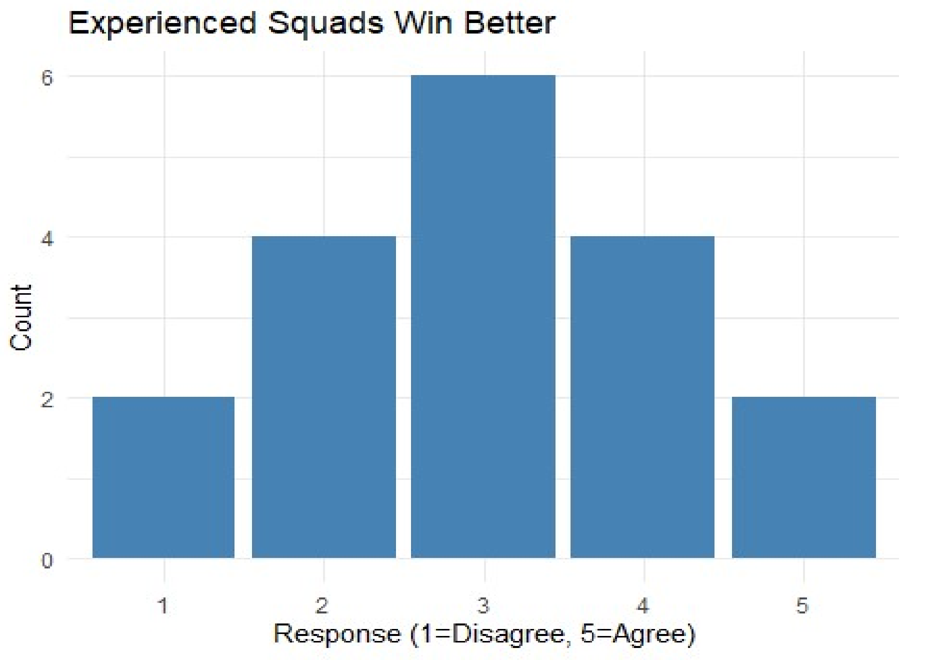



```r
# Chart 2 - Preferred playing style
ggplot(survey, aes(x = play_style, fill = play_style)) +
  geom_bar(show.legend = FALSE) +
  coord_flip() +
  labs(title = "Preferred Playing Style",
       x = "", y = "Count") +
  theme_minimal()
```

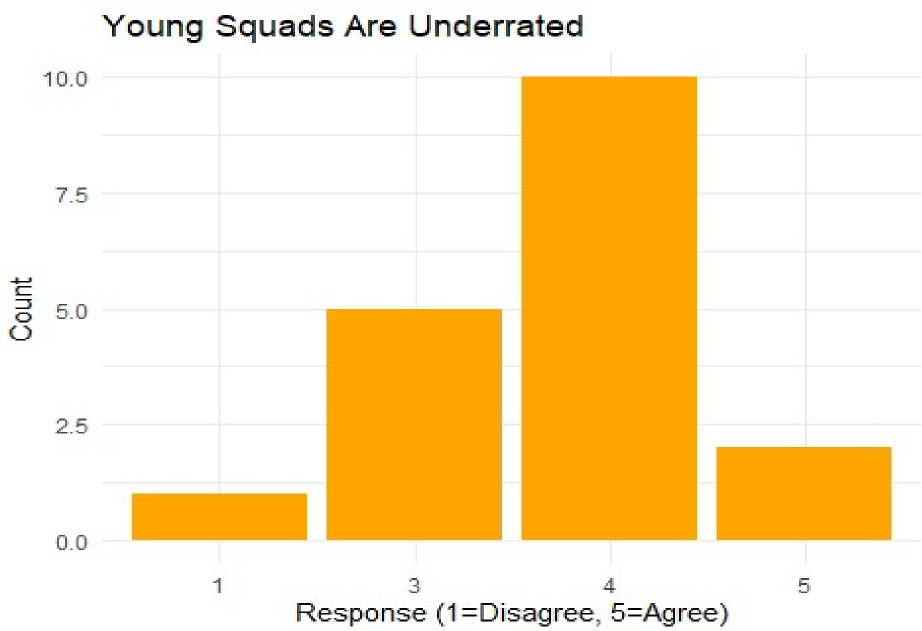

```
# Chart 3 - Strongest region
ggplot(survey, aes(x = best_region, fill = best_region)) +
  geom_bar(show.legend = FALSE) +
  labs(title = "Which Region Produces Strongest Squads?",
       x = "", y = "Count") +
  theme_minimal()
```



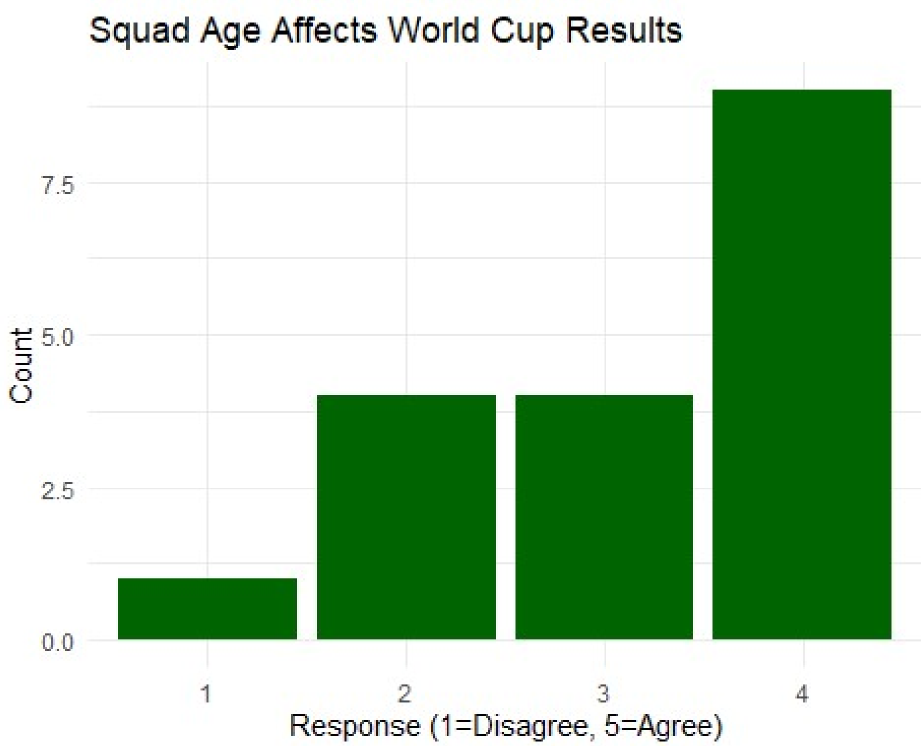
```r
# Experienced squads (older players) perform better at the World Cup?
ggplot(survey, aes(x = factor(experienced_win))) +
  geom_bar(fill = "steelblue") +
  labs(title = "Experienced Squads Win Better",
       x = "Response (1=Disagree, 5=Agree)",
       y = "Count") +
  theme_minimal()
```




```r
# Young squads are underestimated at the World Cup?
ggplot(survey, aes(x = factor(youth_underrated))) +
  geom_bar(fill = "orange") +
  labs(title = "Young Squads Are Underrated",
       x = "Response (1=Disagree, 5=Agree)",
       y = "Count") +
  theme_minimal()
```




```r
# Squad age significantly affects World Cup results?
ggplot(survey, aes(x = factor(age_affects))) +
  geom_bar(fill = "darkgreen") +
  labs(title = "Squad Age Affects World Cup Results",
       x = "Response (1=Disagree, 5=Agree)",
       y = "Count") +
  theme_minimal()
```



Conclusion

Squad age does matter in football, just not in the way most people expect.

The data shows age has a real but weak effect on goals scored. Statistically significant but R squared of 0.009 tells you it's not the main story. Reaching the knockout stage is a different matter though, age has no meaningful effect there at all. So picking an older squad doesn't guarantee you go further in the tournament.

The confederation finding is probably the strongest result. African squads are genuinely and significantly younger than European ones. That's a real pattern across 90 years of data.

The survey was interesting too. Every single respondent picked 27 to 29 as the ideal squad age, which lines up almost perfectly with the actual historical average of 27 years. Fans have a surprisingly accurate intuition about this even without looking at the numbers.

Squads have also gotten older over 90 years of football. From around 25 in 1930 to nearly 28 in 2018. Modern football clearly favors experienced players. But experience alone doesn't win tournaments. Some of the oldest squads in history got knocked out in the group stage.

“Age is a factor, just not the deciding one”